



Virgin Galactic in space

Richard Branson's latest test flight, the Virgin Galactic, has made it to the edge of space and back. <https://dgit.in/jan19-81>



Global quantum communication

Researchers from Italy prove that global quantum communication will be possible between satellites and ground station. <https://dgit.in/jan19-82>

telescope, the Fermi Gamma Ray telescope, and the LIGO collaboration. The orbit@home project lets you participate in planetary defence, by identifying any asteroids that may one day strike the Earth. The Milky-Way@home and Cosmology@Home projects simulate the formation and evolution of galaxies and universes, then compare them to real observed values. These projects are a way for astrophysicists to test their theories. Contributing to distributed computing projects is the simplest way to participate in space science.

NEEDLE IN A HAYSTACK

One of the most rewarding programs for citizen scientists is the Planet Hunters project, which is being followed up with the Planet Hunters TESS project, which launched on 6 December 2018. In the initiative, citizen scientists have to go through the data gathered by the Transiting Exoplanet Survey Satellite (TESS), to identify dips in the brightness of stars. These dips are telltale signs of a transiting exoplanet, or a planet coming in between the star and our Earth. Over the course of the Planet Hunters project, which used data from the Kepler mission, over 2,000 exoplanet candidates were observed. The data from TESS promises to provide many more discoveries.

NASA's Kepler mission, which was one of the most prolific exoplanet hunting telescopes in existence, was helped considerably by citizen scientists pitching in. The participation of amateur astronomers in these programs allows for scientists to rapidly process vast amounts of data. The programs also provide a career path for those participating, and citizen scientists can contribute according to their level of expertise. One of the main advantages of using a pair of human eyes is that if there is something unusual in an image, that will be ignored by machine learning algorithms trained to detect particular phenomena, while a human will still flag it as something unexpected.

HOW TO PARTICIPATE IN COLLABORATIVE SPACE PROJECTS



ZOONIVERSE

This is the absolute best place to find exciting new space projects, and hosts the popular Planet Hunters collaboration. Exoplanets, solar storms, terrains of planets, supernovae... you can study all of these here.

<https://dgit.in/zoospace>

RAD@HOME

An initiative by astronomers from India, allows people from a science background to participate in real research. There are discovery camps, which you have to sign up for, and most of the coordination takes place through Facebook.

<https://dgit.in/rad>

HIWISH

You can take control of the HiRISE camera on board the Mars Reconnaissance Orbiter. You just have to identify a previously unexplored region of Mars, and fill out a form explaining why you would want to take a picture of that particular location.

<https://dgit.in/hiwish>

NASA SOLVE

This is a mailing list that allows you to stay updated on the latest citizen science projects, as and when such efforts are required by the scientific community. By signing up to the mailing list, you become a NASA SOLVE-r.

<https://dgit.in/nasaslv>

The Milky Way Project uses public participation to identify expanding bubbles caused by stellar wind using the data from the Spitzer Space Telescope. Volunteers are first trained on a list of objects to identify, and then mark them out on photos that no one else may have ever seen before. NASA STEREO provides data to the Solar Storm Watch project. Citizen scientists track the progress of solar storms to better improve space weather predictions.

CROWDFUNDING

Apart from computational resources, and sorting images, another way for regular people to participate in space based missions is through crowdfunding. To better understand the mysteries of KIC 8462852, an interesting astronomical object first flagged by citizen scientists, Tabetha Boyajian, an American astrophysicist launched a successful kickstarter campaign to dedicate more time from telescopes towards observing the object. The star was experiencing dips of a degree greater than could be explained by any transiting exoplanet. This led to a number of outlandish theories to explain the cause, including the existence of a superstructure – a dyson ring around the star that would be a sign of an advanced extraterrestrial civilisation. Scientists now think that a ring of debris is the most likely cause of the dips, but since the dips do not follow a regular frequency or intensity, the star does require further study. The Las Cumbres Observatory Global Telescope Network (LCOGT) donated 200 observation hours towards the campaign even before the kickstarter was successful. The project continues to engage with citizen scientists, with each dip in brightness being assigned a particular name.

The Lightsail program is entirely citizen-funded, and has raised over a million US dollars. The program hopes to demonstrate feasible long distance spaceflight using cheap propulsion through solar sails, using nanosatellites of the CubeSat standard. The first prototype mission, the Lightsail, was launched in 2015, and successfully demonstrated the deployment of a solar sail in space. The spacecraft then took images of the Earth. Lightsail 2, will attempt the first solar sail powered orbit around the Earth, and is contracted to launch on a SpaceX Falcon Heavy some time in early 2019. The learnings that emerge from the Lightsail project could allow humans to one day realise cheap interstellar travel. 